

ENGR 498: Interdisciplinary Capstone
University of Arizona

Final Report

Automated Weight-Bearing 3D Ultrasound Foot Scanner

Project Number: 26045

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1 Introduction and Project Description

1.1 Introduction

The arch of the foot supports the body's weight. When the posterior tibial tendon weakens, the plantar fascia stretches and the arch collapses, which can lead to hindfoot valgus and forefoot abduction. This is known as Progressive Collapsing Foot Deformity (PCFD), which affects five million people in the United States yearly, with more than ten percent of adults 65 years and older estimated to have PCFD. Currently, diagnostic tools for PCFD include x-rays, CT scans, and ultrasound. However, PCFD needs fast and accurate diagnosis for proper treatment, and no existing system combines weight-bearing capability with automated, repeatable scanning.

The proposed project aims to address a critical gap in non-invasive diagnostics for foot arch health by automating ultrasound and shear wave elastography (SWE) imaging to visualize and measure stiffness of the soft tissues of the foot under weight bearing conditions. Current methods lack consistency, making it very difficult for clinicians to detect early signs of arch collapse. SWE can detect the smallest changes in tissue stiffness and, with proper integration, it has the ability to be a major advancement in preventative care for conditions associated with arch collapse, such as plantar fasciitis and other musculoskeletal disorders. B-mode imaging, commonly used in sonograms, can provide a depth based image of the scanned tissue. This project uses B-mode in congruence with SWE to help the clinician identify that they're scanning the desired tissue.

1.2 Project Description

The University of Arizona Department of Biomedical Engineering has requested project improvements of the 3D ultrasound foot scanner to include automatization of data acquisition, refining the data processing software, and the graphical interface of the design. Improvements will include a fully functioning 2D liner array to control the scanning, a GUI interface to modify scan parameters, wiring clean up, and improved usability in the clinical and laboratory settings.

Previous years laid the mechanical foundation for the project, including the weight-bearing platform and linear actuator. The scope this year focuses on the full volumetric reconstruction of the soft tissues for detailed analysis. Our goal is to integrate the mechanical foundation with accurate ultrasound data acquisition and analysis. We aim for the project to be user-friendly in clinical environments, adaptable with different foot sizes, and effective for clinical assessments of foot pathology. The scanner will be deployed in clinical and laboratory settings to detect early signs of arch collapse. During use, the patient will stand on the platform as the system captures B-mode and SWE data along the indicated area. Data collected by the device will be processed and visualized through an intuitive graphical interface, enabling clinicians to analyze results quickly and make informed decisions regarding treatment or prevention.

2 System Description

2.1 System Description

The Automated Weight-Bearing 3D Ultrasound Foot Scanner is a medical imaging device designed to measure the stiffness of arch-supporting structures in the foot while the patient is standing. Most imaging systems are not performed during active contractions of tendons and ligaments happening while a patient stands, allowing measurements to be more representative of everyday load bearing conditions. The device is intended to support clinicians in identifying and evaluating conditions like plantar fasciitis and arch collapse.

As shown in figure DWG-002 (found in the next section), the foot scanner is composed of four main subsystems.

1. Mechanical Assembly: Provides a rigid, weight-bearing platform for the patient and houses the motorized probe carriage. The assembly positions and translates the ultrasound probe along the plantar surface of the foot, and a coupling pad maintains consistent acoustic contact throughout the scan.
2. Power Subsystem: supplies regulated power to all the mechanical, ultrasound, and computational components, ensuring stable operation during scanning.
3. Ultrasound Subsystem: acquires both B-mode images for anatomical mapping and shear-wave elastography data for quantitative tissue-stiffness analysis. This subsystem interfaces directly with the probe mount and synchronizes image capture with probe motion.
4. Software Subsystem: controls the motor actuators, image acquisition timing, and data processing through a graphical user interface. The software manages scan execution, displays intermediate and final results, and processes raw data in MATLAB to generate stiffness maps, which are then stored in organized directories for later review.

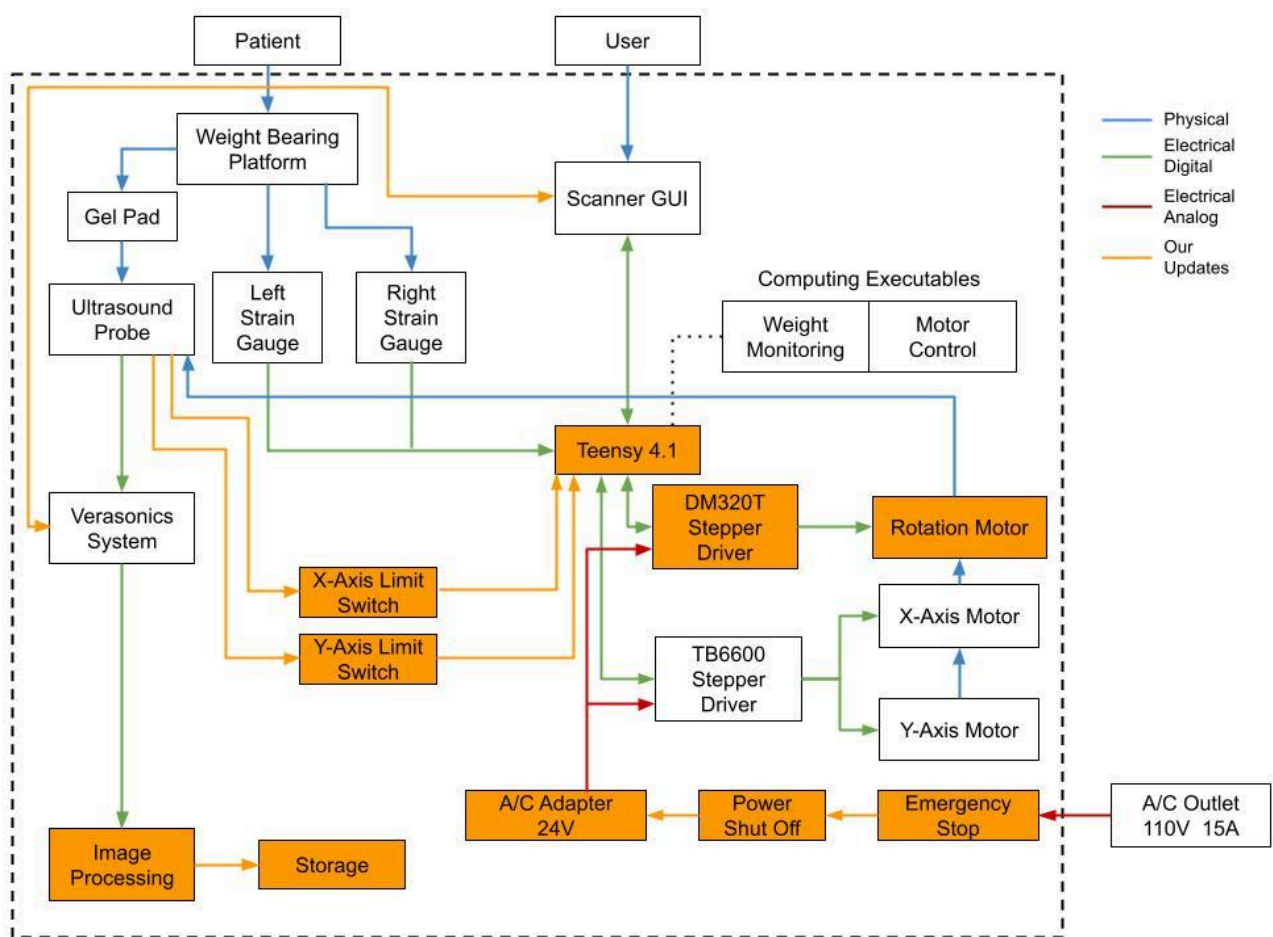
The minimum viable products established the foundational capabilities of the system and expanded its functional scope.

- MVP1 focused on implementing the automated-scanning workflow by developing a graphical user interface that coordinated motion control with the Verasonics platform, enabling synchronized image acquisition through the scan sequence. This phase also supported full volumetric reconstruction of soft tissue and incorporated an interchangeable heel cup to accommodate different foot shapes.
- MVP2 advances the system toward more controlled and clinically relevant imaging by determining the optimal step size for accurate three-dimensional reconstruction, evaluating both vertical and horizontal probe orientations for improved image quality, and generating a 3D stiffness map to support the elastography analysis.

The goal of this product is to provide an intuitive, comprehensive, and efficient method for clinicians to perform ultrasound scans of the foot and quickly receive a volumetric image of

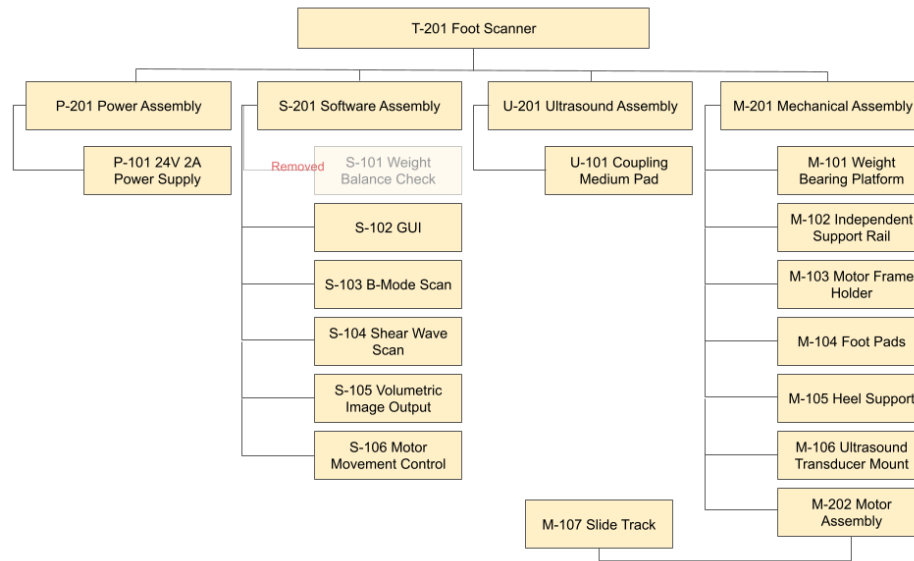
said foot. This can be useful for patients and clinicians globally, since the conditions the system is checking for are not unique to our current location. Our product's ability to accommodate a wide range of foot sizes allows use across a diverse patient population. This system provides a noninvasive, quick weight-bearing assessment of foot stiffness, which will help improve early diagnosis and treatment planning for conditions like PCFD. The product will also aid the clinician by providing stiffness analysis, enabling better decision making towards the patient's condition. In general, our product is intended to support a broad range of patients and be useful in the orthopedic care setting due to the practicality, accessibility, and ease of use of the system.

2.2 System Block Diagram



DWG-001 System Block Diagram

2.3 Subsystems



DWG-002 Subsystem

3 Technical Data Package

3.1 Introduction

Refer to [section 1.0](#) for information regarding the introduction and project description.

3.2 System Description

Refer to [section 2.0](#) for information regarding the system description and system block diagram.

3.3 System Verification Plan

The System Verification Plan (SVP) defines the methods used to verify that MVP1 and MVP2 meet the system requirements by specifying the verification methods, procedures, and support analyses applied through testing. Each requirement is assigned to a verification method: inspection, analysis, test or demonstration, with requirement-to-method traceability documented in the system requirement verification matrix (SRVM) table in [Section 3.4.2.5](#). Detailed verification procedures for each requirement are provided in [Appendix A](#), and expected system performance is supported by analytical and computational models referenced in [Section 4](#). Together, these elements establish the verification framework for the MVP and confirm compliance with the defined system requirements.

3.4 Design Documentation

The Design Documentation section presents the technical materials that define the current MVP, including the Item Drawing List (IDL), system requirements, verification documentation, hardware drawings, software documentation, and supporting models. This section captures the artifacts that describe the system's mechanical, electrical, and software elements, and provides the formal basis for fabrication, integration, and verification activities.

3.4.1 Item Drawing/Document List

Drawing Number	Title	Revision	Status
System-Level Documentation			
DWG-001	System Block Diagram	D	Under Revision
DWG-002	Subsystems	B	Released
Mechanical System			
DWG-004	Array Mount	D	Released
DWG-005	Heel Cup	F	In Progress
DWG-006	Y-Axis Linear Actuator	N/A	Integrated
DWG-007	X-Axis Linear Actuator	N/A	Integrated
DWG-008	Limit Switch Mount	D	Released
DWG-009	Probe Mount	A	Released
DWG-010	Rotation Motor Mount	D	Released
DWG-011	Gel Pad Holder	C	Released
Electrical/Power System			
DWG-012	Wiring Diagram	A	In Progress
DWG-013	Teensy 4.1 Pinout	N/A	Integrated
Ultrasound System			
US-001	Aquaflex Gel Pad	N/A	Integrated
US-002	GE 9L-D Transducer	N/A	Integrated
Software System			
DWG-003	System Design Document	B	Released
SW-001	MATLAB Motor Control Function*	D	Released
SW-002	B-mode MATLAB Script*	F	In Progress
SW-003	SWE MATLAB Script*	Pending	Pending

SW-004	Verasonics GUI	C	In Progress
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**All MATLAB scripts are stored in the project GitHub repository:*

<https://github.com/ruthsalazar/Automated-Weight-Bearing-Ultrasound-Foot-Scanner>

3.4.2 System Requirements Document

3.4.2.1 Introduction and Scope

Refer to sections 1.0 and 2.1 for the introduction and system description.

3.4.2.2 Applicable Documents

OSHA 1910.29(b)(1): Fall protection systems and falling object protection - criteria and practices. <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.29>

IBC 1011.5.2: Riser height and tread depth.

<https://codes.iccsafe.org/s/IBC2018/chapter-10-means-of-egress/IBC2018-Ch10-Sec1011.5.2>

ASTM D638 – 14: Standard Test Method for Tensile Properties of Plastics:

<https://compass.astm.org/content-access?contentCode=ASTM%7CD0638-14%7Cen-US>

ANSI C84.1-2006: American National Standard for Electric Power Systems and Equipment -- Voltage Ratings (60 Hertz) 4.2.5 The heel cup shall accommodate foot width up to 10 cm.

https://durastudio.com/sites/default/files/ansi_c84-1-20xx_2016_revision_draft_2019-02-15.pdf

IEC 62359: Ultrasonics – Field Characterization – Test Methods for the Determination of Thermal and Mechanical Indices Related to Medical Diagnostic Ultrasonic Fields.

<https://cdn.standards.iteh.ai/samples/16301/3acce1308e2400098e170e515757a75/IEC-62359-2010.pdf>

Texas Instruments: Signal Processing Overview of Ultrasound Systems for Medical Imaging.

https://www.k-space.org/ymk/sprab12.pdf?utm_source

Three-Dimensional Shear Wave Elastography Using a 2D Row Column Addressing (RCA) Array:

<https://pmc.ncbi.nlm.nih.gov/articles/PMC10521701/>

Ultrasonic Shear Wave Elasticity Imaging Sequencing and Data Processing Using a Verasonics Research Scanner.

<https://verasonics.com/wp-content/uploads/2017/10/Deng-2017-UFFC-Ultrasonic-Shear-Wave-Elasticity-Imaging-Sequencing-and-Data-Processing-Using-a-Verasonics-Research-Scanner.pdf>

Elsevier: Ultrasound Signal Processing: From Models to Deep Learning

<https://pdf.sciencedirectassets.com>

Aquaflex Ultrasound Gel Pad Datasheet:

<https://cdn.shopify.com/s/files/1/1046/1086/files/Parker-Aquaflex-Ultrasound-Gel-Pad-Safety-Data-Sheet.pdf>

GE 9L-D Transducer Data for Verasonics Vantage System:

<https://verasonics.com/ge-transducers-for-vantage-systems/>

Definitions, Acronyms, and Abbreviations

The following nomenclature is used throughout the System Requirements Document.

Term	Description
PCFD	Progressive collapsing foot deformity.
OSHA	Occupational Safety and Health Administration
IBC	International Building Code
ASTM	American Society for Testing and Materials
ANSI	American National Standards Institute
IEC	International Electrotechnical Commission

3.4.2.3 System Requirements

The system requirements are statements that act as design constraints and guidelines for the development of the system. They aim to ensure the system is satisfactory to the customer's needs and compliant with regulatory standards.

3.4.2.3.1 Performance Requirements	The scanner shall generate 3D volumetric images with spatial resolution ≤ 0.10 mm. Shear wave velocity measurements shall have accuracy within $\pm 15\%$. The system shall operate under weight-bearing conditions up to 120 kg. The system shall have a total set up time of less than 10 minutes.
3.4.2.3.2 Characteristic Requirements	The system shall include a heel support that enables imaging of the foot arch without movement The heel cup shall ensure durability, patient comfort, and biocompatibility. The system shall accommodate foot sizes up to men's 14. The heel cup shall accommodate foot width up to 10 cm.
3.4.2.3.3 Interface Requirements	The system shall provide two UI modes: Simple and Advanced. The user interface shall display real-time 2D ultrasound images during scanning. The system shall export imaging data in DICOM format for compatibility with external analysis software. The system shall include USB 3.0 and Ethernet ports for data transfer and device connectivity.
3.4.2.3.4 Safety	The system shall comply with FDA ultrasound exposure limits. The system shall include an emergency stop function accessible to the

Requirements	operator. The system shall include electrical isolation between the power supply and the patient-contact components.
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3.4.2.4 Verification Requirements

The chart below describes the methodology used to verify the system requirements.

3.4.2.4.1 Performance Requirements	An analysis shall be performed showing that the scanner generates 3D volumetric images with spatial resolution ≤ 0.10 mm. An analysis shall be performed showing that the shear wave velocity measurements are within $\pm 15\%$. It shall be tested that the system can operate under weight-bearing conditions up to 120 kg. A demonstration shall be used to verify a total set up time of less than 10 minutes.
3.4.2.4.2 Characteristic Requirements	It shall be demonstrated that the system includes a heel support that enables imaging of the foot arch without movement An inspection shall take place to verify that the heel support is fabricated to ensure durability, patient comfort, and biocompatibility. An inspection shall take place to verify that the system can accommodate foot sizes up to men's 14. An inspection shall take place to verify that the system can accommodate foot width up to 10 cm.
3.4.2.4.3 Interface Requirements	It shall be verified that the system has 2 UI modes: Simple (pre-set parameters) and Advanced (expanded controls and preferences). It shall be demonstrated that the user interface displays real-time 2D ultrasound images during scanning. It shall be tested that the system can export imaging data in DICOM format for compatibility with external analysis software. It shall be confirmed through inspection that the system includes both USB 3.0 and Ethernet ports for data transfer and device connectivity. Successful file transfer shall be shown through both ports.
3.4.2.4.4 Safety Requirements	It shall be verified with testing and inspection that the system complies with FDA ultrasound exposure limits. Compliance shall be shown with both test data and reports. It shall be demonstrated that the system includes a stop function accessible to the operator. This stop function shall immediately stop the scan. It shall be verified with safety testing that there is electrical isolation between the power supply and the patient-contact components.

3.4.2.5 System Requirements Verification Matrix

The Systems Requirements Verification Matrix outlines the methods of verifying the system before production. The following verification methods are assigned in the matrix below:

T – Test: Interfacing with the system to extract data which proves functionalities.

A – Analyze: Technical or mathematical models, simulations, or scientific procedures or principles which prove functionalities.

D – Demonstrate: Simple operation of the system to show functionality or characteristics.

I – Inspect: simple physical manipulation or measurements to show requirement compliance.

Performance Requirements

Requirements	Verification Method			
	T	A	D	I
The scanner shall generate 3D volumetric images with spatial resolution ≤ 0.10 mm.		x		
Shear wave velocity measurements shall have accuracy within $\pm 15\%$.		x		
The system shall operate under weight-bearing conditions up to 120 kg.	x			
The system shall have a total set up time of less than 10 minutes.			x	

Characteristic Requirements

Requirements	Verification Method			
	T	A	D	I
The scanner shall accommodate foot sizes up to men's 14.				x
The heel cup shall be fabricated with biocompatible materials.				x
The heel cup shall accommodate foot width up to 10 cm.				x

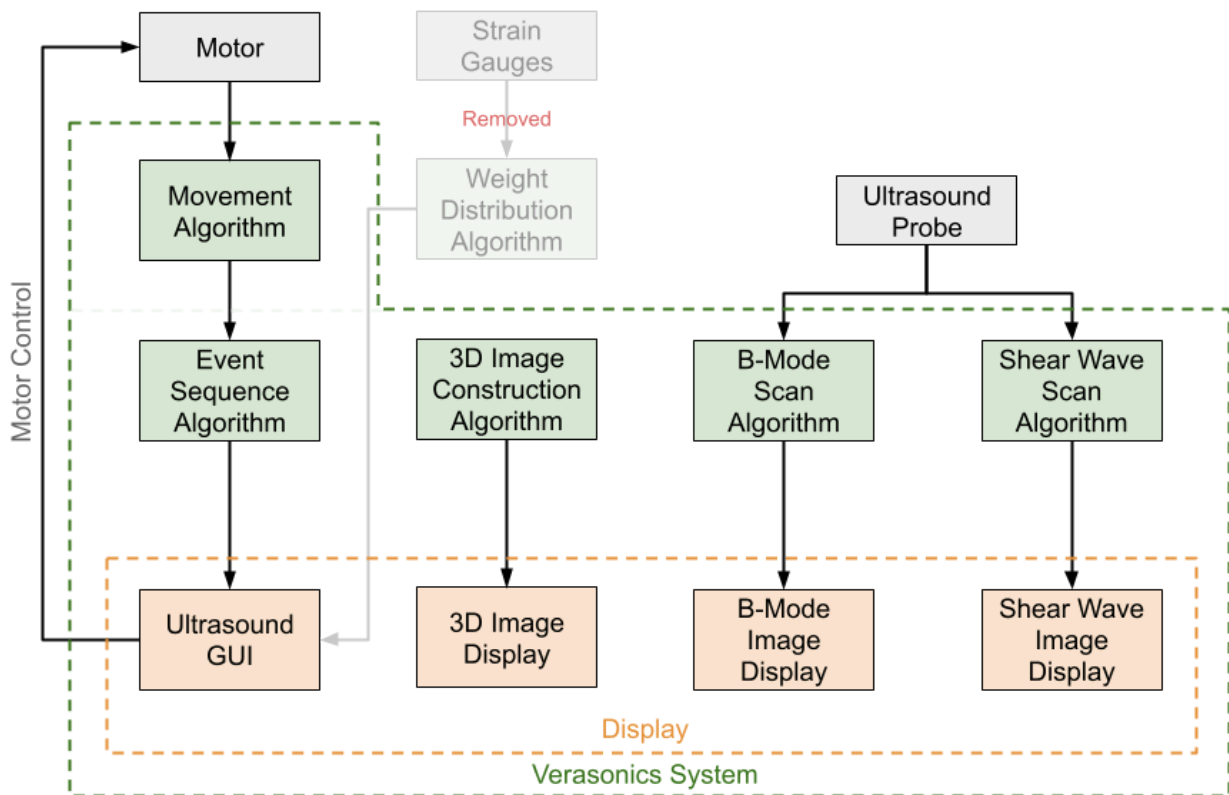
Interface Requirements

Requirements	Verification Method			
	T	A	D	I
The system shall display real-time ultrasound images during scanning.			x	
The system shall store scan data in DICOM format.	x			
The system shall provide a USB 3.0 connection for data transfer.				x

Safety Requirements

Requirements	Verification Method			
	T	A	D	I
The system shall comply with FDA ultrasound exposure limits.	x			x
The system shall include a stop function accessible to the operator.			x	

System Design Document



DWG-003 System Design Document

3.4.3 Verification Documentation

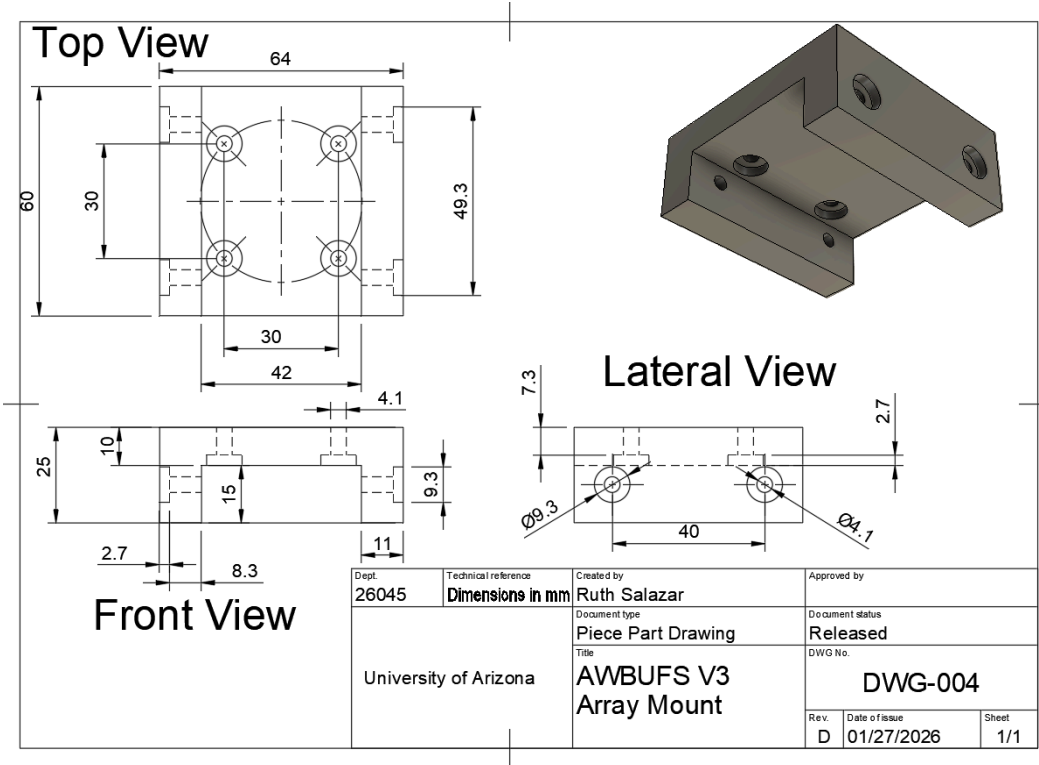
Refer to [Section 5](#) and [Appendix A](#) Verification Documentation.

3.4.4 Hardware Drawing Package

3.4.4.1 Piece-Part Drawings

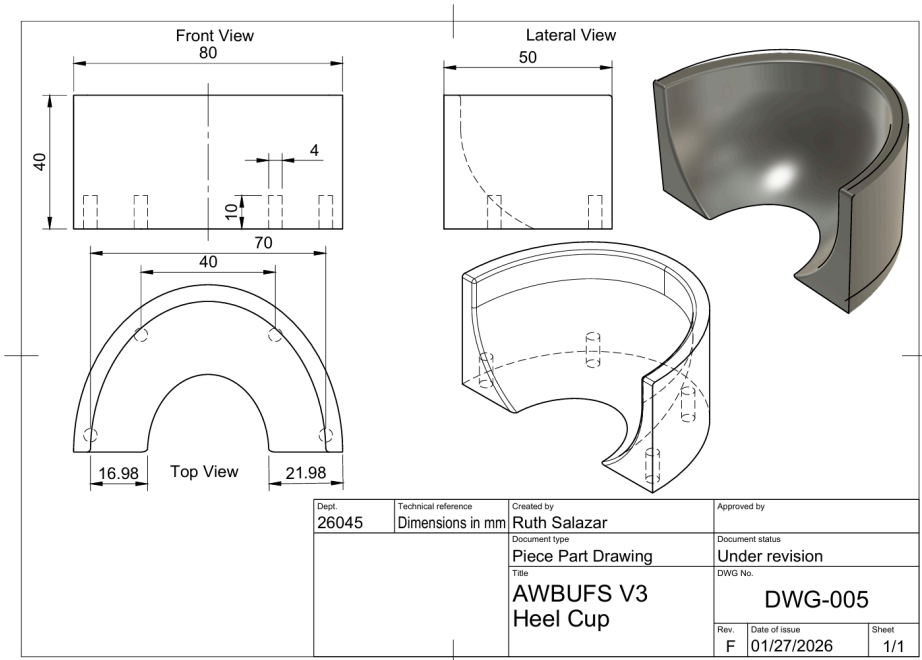
For the following drawings, a revision for precision tolerances will be implemented to communicate to manufacturers important threshold limits. Once designs are finalized, specifically for those still “in-progress,” drawings will be up to date.

X & Y-axis Array Mount



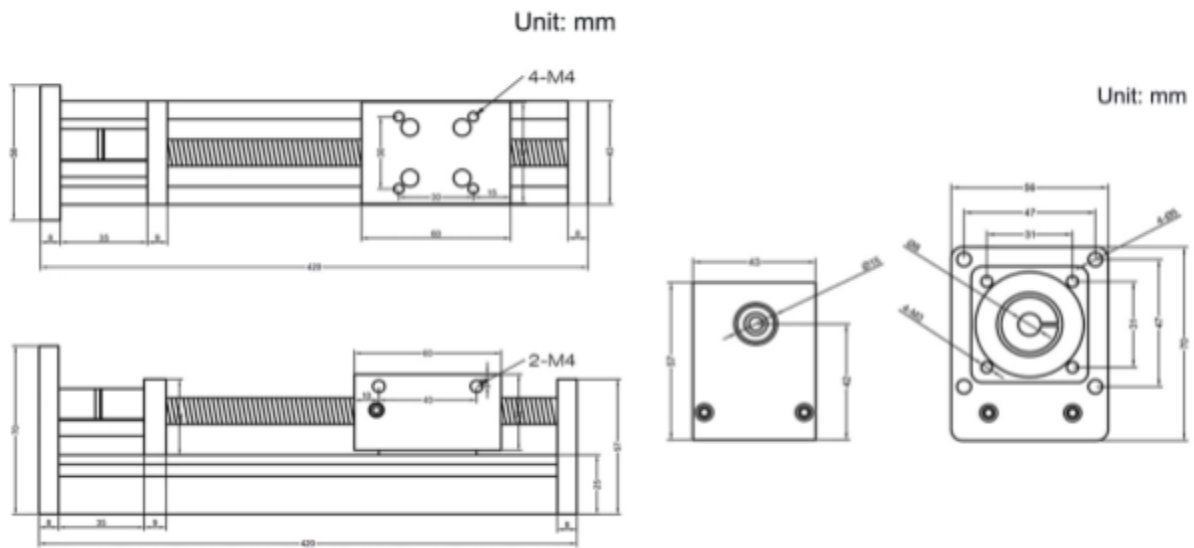
DWG-004 Array Mount

Heel Cup

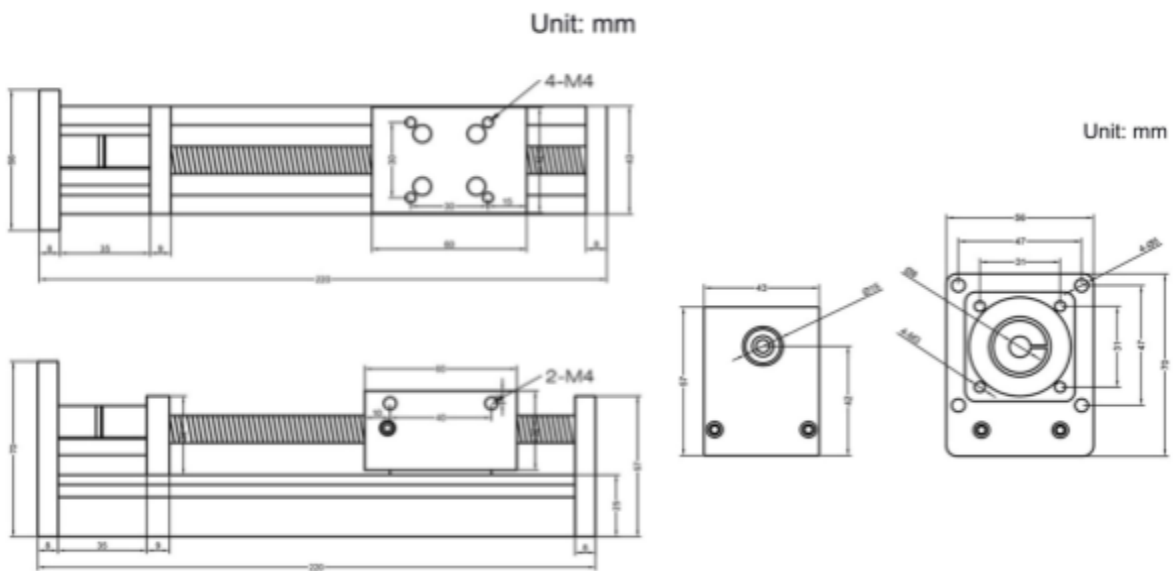


DWG-005 Heel Cup

Linear Actuators

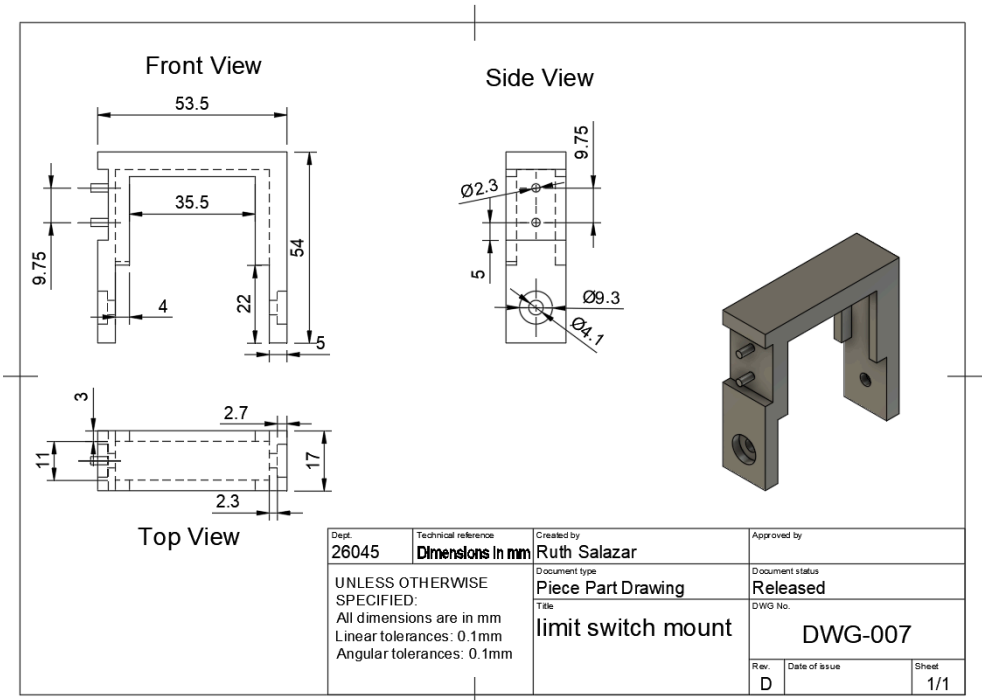


*DWG-006 Y-Axis Linear Actuator
Drawing obtained from datasheet.*



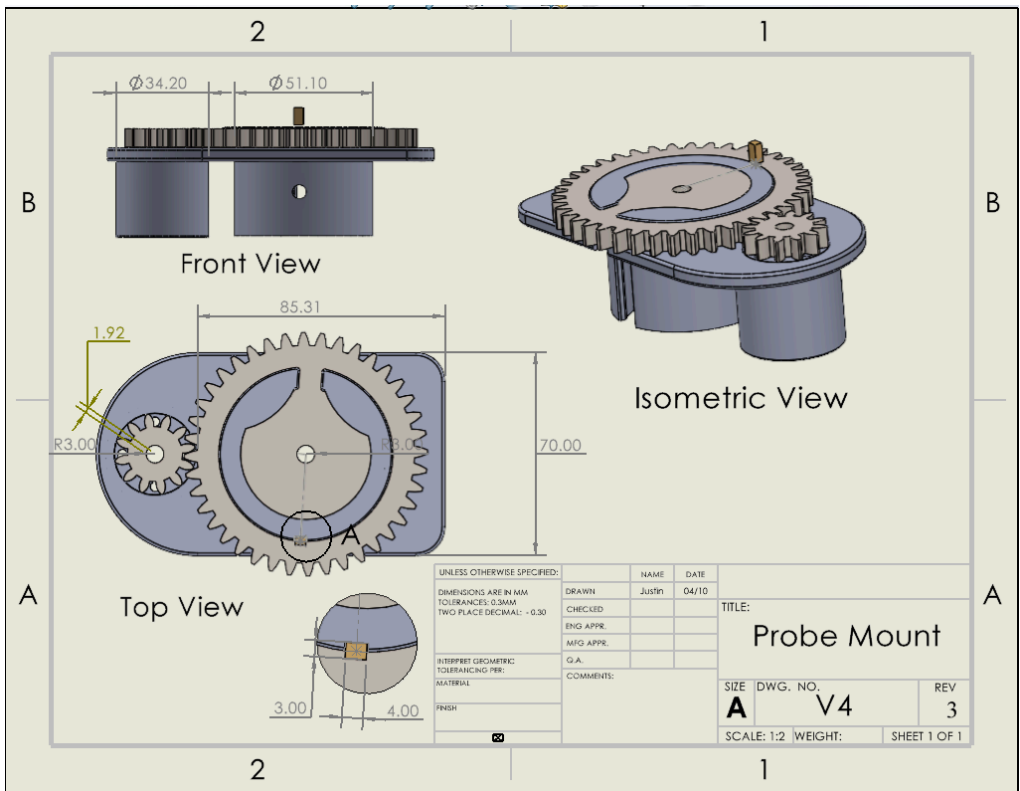
*DWG-007 X-Axis Linear Actuator
Drawing obtained from datasheet.*

Limit Switch Mounts



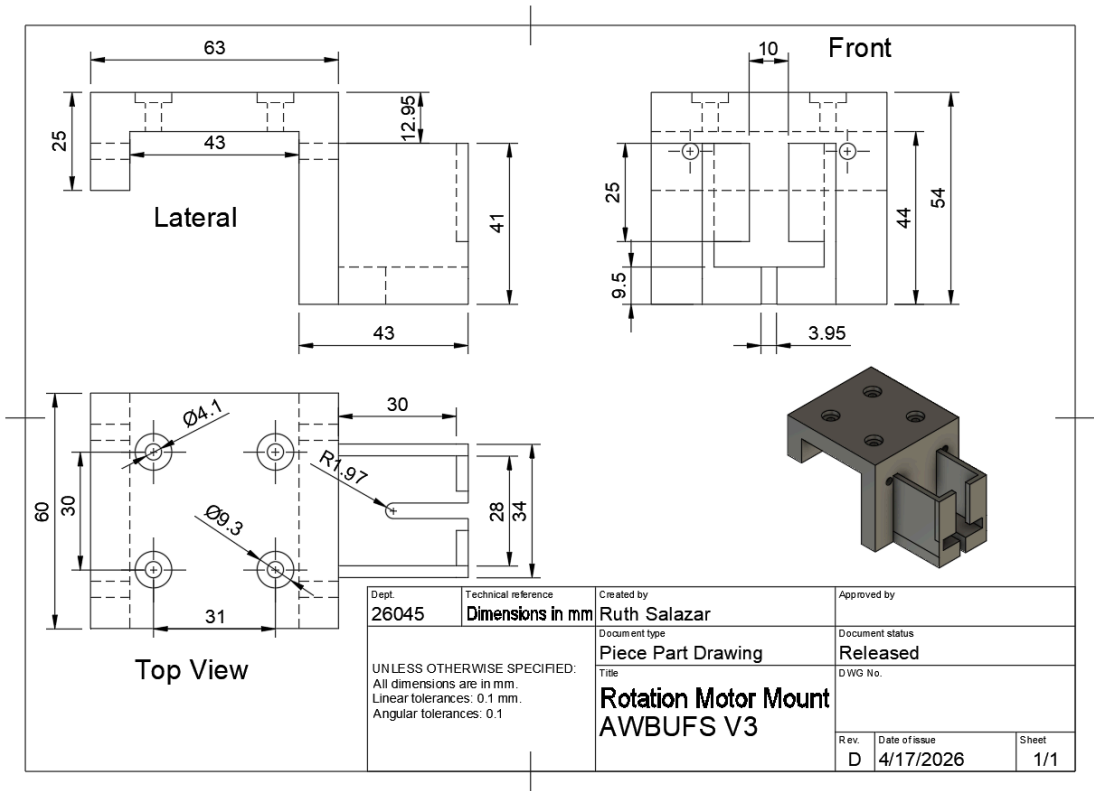
DWG-008 Limit Switch Mount

Probe Mount



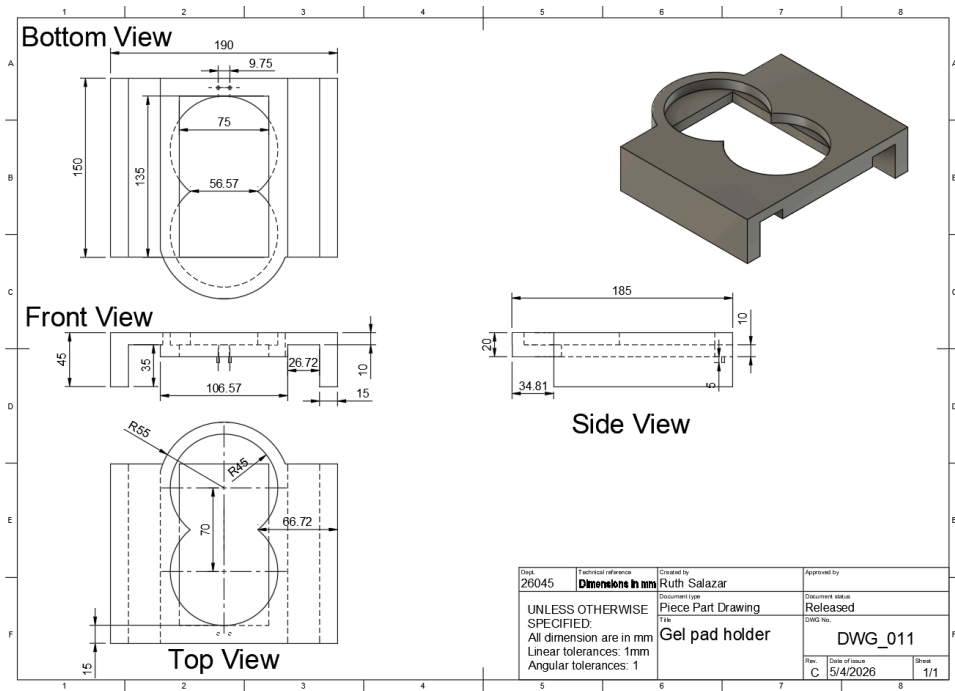
DWG-009 Probe Mount

Rotation Motor Mount



DWG-010 Rotation Motor Mount

Gel Pad Holder



DWG-011 Gel Pad Holder

3.4.4.2 Ultrasound

Gel Pad



*US-001 Aquaflex Gel Pad
Image obtained from datasheet*

Transducer



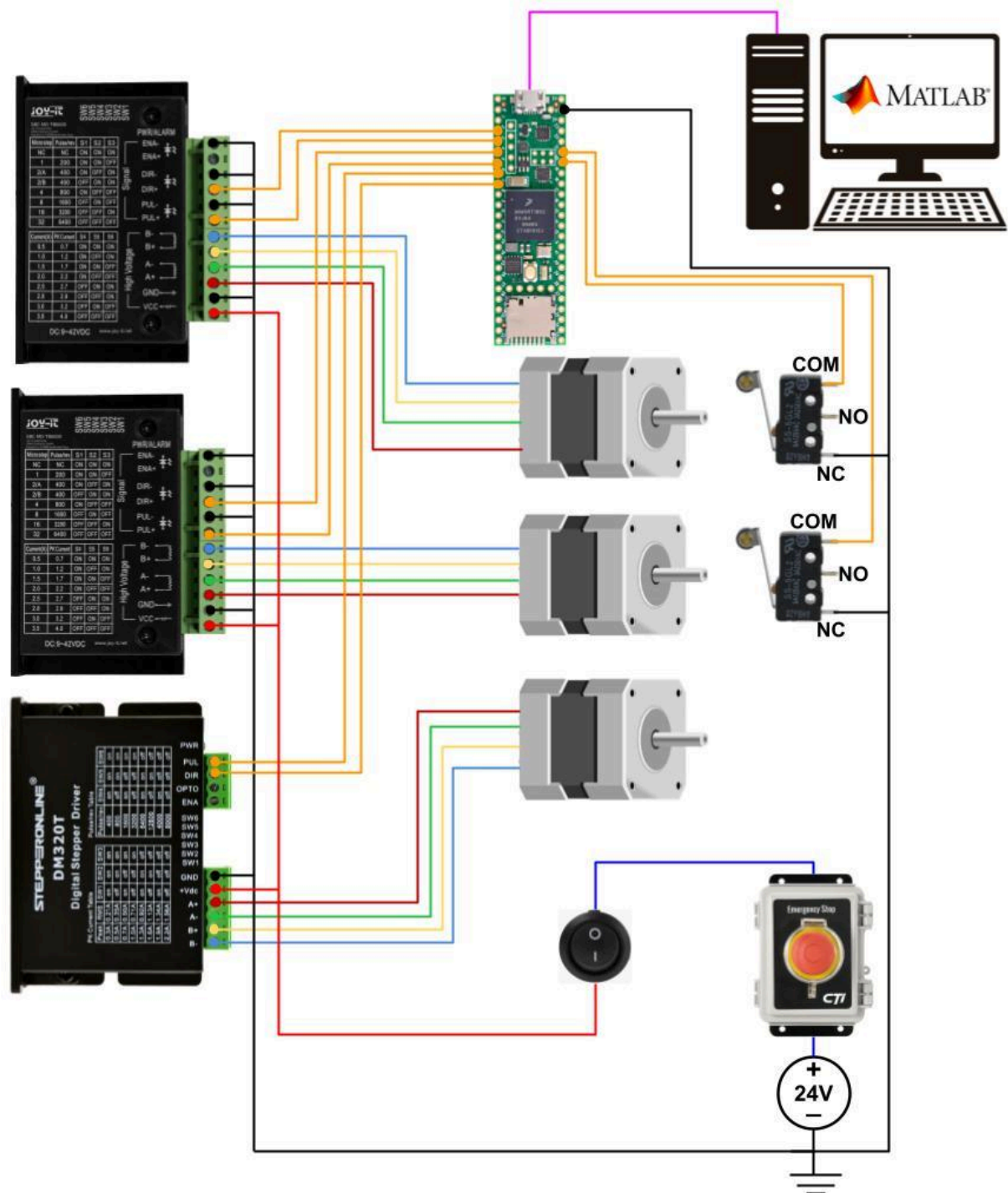
9L-D

Center Frequency	5.3 MHz
Bandwidth	75%
Number of elements	192
Pitch	0.23 mm
Elevation Focus	28 mm
Elevation Aperture	6.0 mm
Mux	No
Compatibility	Vantage 256

*US-002 GE 9L-D Transducer
Image obtained from Verasonics website.*

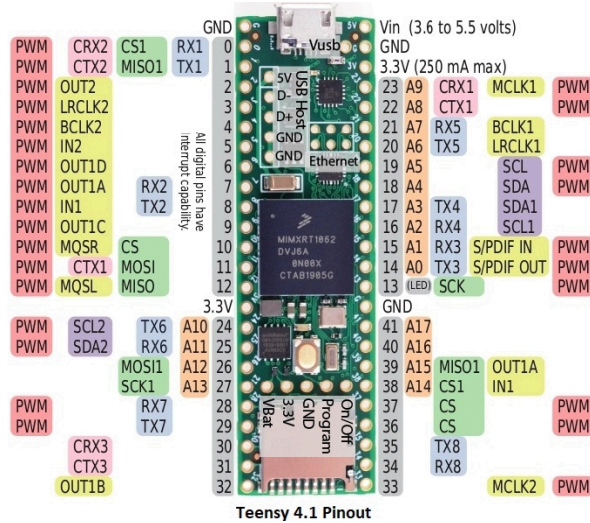
3.4.4.3 Electrical Drawings

Wiring Diagram



DWG-012 Wiring Diagram

Teensy 4.1 Board Layout

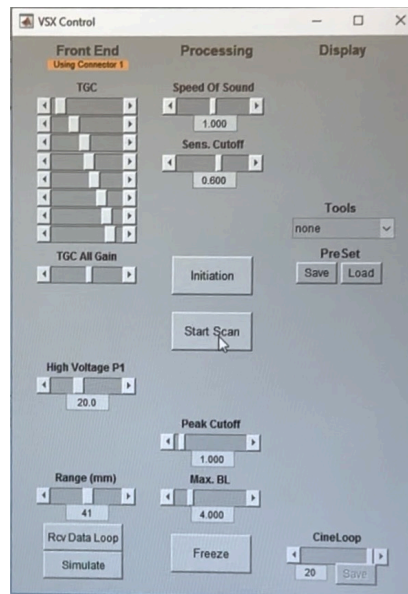


DWG-013 Teensy 4.1 Pinout

Drawing obtained from datasheet.

3.4.5 Software Documentation

The software design documentation (SDD) outlines the software design for the project. This document provides a comprehensive roadmap for the software's role in achieving project goals and addressing technical and operational requirements. The SDD outlines the scope, critical design decisions, and the implementation of B-mode software and SWE that is supported by a robust (GUI). Additional information regarding the data handling, ensuring compliance with relevant standards, and a requirement traceability matrix linking software to specific objectives are also included in the software design documentation. The current revision of the SDD can be found in Appendix B.



SW-004 Verasonics GUI

3.4.6 Models

Refer to [Section 4](#) for models and analysis.

4 Models/Analyses

The system models developed for MVP1 establish the analytical performance expectations of the automated scanning platform, positioning accuracy, and timing behavior. The motion model predicts a linear positioning resolution of 0.5 mm per step, with expected linear-axis accuracy of ± 0.2 mm and repeatability of ± 0.25 mm after accounting for mechanical compliance and backlash. The rotation axis is modeled to achieve $\pm 3^\circ$ accuracy and $\pm 1^\circ$ repeatability across its full range of motion. Timing models based on commanded step rates and communication overhead predict a total scan time of approximately 120 s, with a tolerance of ± 10 s, assuming uninterrupted motion and ideal synchronization. A linear calibration model for the weight-bearing platform predicts $\pm 2\%$ accuracy, $\pm 1\%$ repeatability, and linearity of $R^2 \geq 0.98$, supporting consistent load distribution during imaging.

MVP2 expands the modeling framework to focus on image quality and reconstruction performance, reflecting the transition from basic automation to clinically meaningful imaging. Step-size modeling predicts that spatial sampling must satisfy the Nyquist criterion for the ultrasound system's lateral resolution, with an expected optimal step size in the range of 0.25-0.50 mm to avoid aliasing and maintain continuity in the reconstructed volume. Probe-orientation models evaluate vertical versus horizontal alignment by predicting differences in acoustic coupling stability, field-of-view coverage, and shear-wave propagation paths, with the expectation that vertical orientation maximizes contact uniformity while horizontal orientation increases lateral coverage. Stiffness-mapping models assume a uniform reconstruction grid and predict that reliable 3D stiffness maps require sufficient shear-wave signal-to-noise ratio, consistent probe-tissue coupling, and a step size small enough to preserve spatial gradients in tissue stiffness. These predictive models define the expected imaging performance of MVP2 in the absence of empirical testing and guide the selection of scanning parameters for future validation.

5 Verification Results

The automatic weight-bearing ultrasound foot scanner was verified based on interface, characteristic, safety, and performance requirements. Verification's main focus was on confirming that the system could perform the automated motion, synchronized ultrasound scanning, and overall support 3D volumetric image reconstruction and stiffness mapping. The verification was completed according to the System Requirement Verification Table using either analysis, demonstration, testing, or inspection methods.

Requirement	Method	Limit/ Reference	Measured/ Ref Value	Date Completed	Pass/ Fail
1. Interface Requirements	-	-	-		-
a. The system shall provide two UI modes: Default and Advanced.	A	Y/N	No	5/4/2026	Fail
b. The user interface shall display real-time 2D ultrasound images during scanning.	D	Y/N			Pass
c. The system shall export imaging data in DICOM format for compatibility with external analysis software.	T	Y/N	No	5/4/2026	Fail
d. The Verasonics system shall communicate with the Teensy controller via serial connection and trigger motion commands.	T	DWG-006	DWG-006		Pass
2. Characteristic Requirements	-	-	-		
a. The system shall include a heel support that enables imaging of the foot arch without movement.	D	DWG-005	DWG-005		Pass
b. The heel cup shall ensure durability, patient comfort, and biocompatibility.	I	DWG-005	DWG-005		Pass
c. The system shall accommodate foot sizes up to men's 14.	I	≤ 14	≤ 14		Pass
d. The heel cup shall accommodate foot width up to 10 cm.	I	DWG-005	DWG-005		Pass
e. The system shall provide at least 100 mm of Y-axis travel.	I	≥ 100 mm	≥ 100 mm		Pass
3. Safety Requirements	-	-	-		
a. The system shall comply with FDA ultrasound exposure limits.	T,I	IEC 62359	Unable to test		Fail
b. The system shall include an emergency stop function that halts all motion.	T	1 second	1 second		Pass
c. The system shall include electrical isolation between the power supply and the patient-contact components.	T	Data Sheet	Data Sheet		Pass
d. The system shall enforce travel limits on all axes.	D	Y/N	Only Y-Axis		Fail
4. Performance Requirements					
a. The scanner shall generate 3D volumetric images from the sequential 2D ultrasound slices using the commanded slice thickness.	D	Y/N	Only volumetric images		Fail
b. The system shall compute shear wave velocity using the implemented SWE equations, producing consistent results across repeated acquisitions.	A	Y/N	Could not validate the results		Fail
c. The system shall operate under weight-bearing conditions up to 120 kg.	T	<120			Pass
d. The system shall have a total set up time of less than 10 minutes.	D	<10 min	3 min		Pass
e. Step size accuracy shall be within ±0.2 mm over 50 mm travel.	T	±0.2 mm	0.04 mm		Pass
f. The commanded rotation angle shall match measured angle within ±3° over a full 360° rotation.	T	±3°	1°		Pass

6 Final Budget

Starting Budget:			\$4500	
Item:	Item Cost:	Use Tax Cost:	Total Cost:	Status:
ELECTRICAL				
Teensy Microcontroller	\$39.11	\$2.19	\$41.30	Received
256GB Flashdrive & On/Off Switch (placed on same order)	\$28.19	N/A	\$28.19	Received
Emergency Stop Button	\$9.77	N/A	\$9.77	Received
AC/DC Adapter & Limit Switches (placed on same order)	\$138.02	N/A	\$138.02	Received
Stepper Driver and Stepper Gearbox (placed on same order)	\$76.17	\$3.37	\$79.54	Received
Circuit Cable and Barrier Strip (placed on same order)	\$28.15	N/A	\$28.15	Received
Prototype PCB Boards & Terminal Blocks (placed on same order)	\$14.77	N/A	\$14.77	Received
10' USB-A to Micro USB Cable	\$7.82	N/A	\$7.82	Received
MECHANICAL				
1 Sheet Black Acrylic 3mm	\$21	N/A	\$21	Received
COUPLING MEDIUM				
Aquaflex Ultrasound Gel Pad	\$56.83	N/A	\$56.83	Received
MISCELLANEOUS:				

Team Polos	\$23.58 x6	N/A	\$141.46	Received
Design Day Poster	\$75	N/A	\$75	Received
TOTAL:			\$641.85	
REMAINING BUDGET:			\$3,858.15	

7 Lessons Learned and Recommended Next Steps

7.1 Lessons Learned

The development of the ultrasound-based scanning system revealed several technical, logistical, and organizational challenges that influenced both system design and overall project execution. Early integration efforts focused on building a custom external graphical user interface (GUI) to simplify parameter control; however, this approach conflicted with the proprietary Verasonics environment and introduced unnecessary complexity. Transitioning to the native Verasonics interface proved to be a more effective solution, showing that working within existing platform constraints is often more practical than attempting full external integration. The project was also shaped by a learning curve associated with the Verasonics system and MATLAB-based workflows. With limited prior experience, initial progress was slower as the team developed an understanding of system architecture, scripting, and data acquisition. Although this required additional time upfront, it ultimately allowed for more effective system control and modification. This suggests that allocating time early for technical familiarization is important when working with specialized systems.

Logistical factors further impacted development. The mechanical and ultrasound components were located in separate labs, and access to these spaces was limited. This separation disrupted workflow continuity and delayed system integration. As a result, it became clear that early coordination of physical resources, such as securing shared lab space and consistent access to equipment, is important for maintaining steady progress. Project timelines were also influenced by dependencies on external stakeholders, including graduate supervisors and lab personnel who controlled system access. Scheduling conflicts reduced the number of available testing windows and slowed iteration cycles. This highlights the importance of planning around external constraints and maintaining clear communication to better align development activities with resource availability.

From a team perspective, loosely defined roles early in the project led to uneven workload distribution and reduced accountability. Shifting to more clearly defined responsibilities improved efficiency and ensured that tasks were more evenly distributed. This demonstrates the value of assigning ownership within a team, especially for complex,

multidisciplinary projects. Delays in accessing documentation and hardware at the beginning of the project also affected progress. These delays showed that administrative and setup tasks, such as obtaining resources and reviewing prior work, should be prioritized early to avoid slowing down technical development later. In addition, the project emphasized the importance of maintaining balanced priorities. At one stage, the team focused heavily on motor integration while other areas, such as data processing and reconstruction, were underdeveloped. This created challenges later in the workflow. Moving forward, developing different parts of the system in parallel and maintaining awareness of overall project needs can help prevent similar issues. Overall, the project reinforced the importance of working within system constraints, planning for resource limitations, and maintaining clear team structure. Iterative testing, flexibility in adapting to challenges, and consistent communication were all important in progressing toward a functional system. These lessons provide guidance for improving efficiency and coordination in future projects involving complex system integration.

7.2 Recommended Next Steps

Future work should move from developing a functional prototype to transitioning to a more portable and clinically applicable imaging system. The following recommendations show the vital areas to focus on:

3D Volumetric Reconstruction: One of the major next steps will be creating the algorithms for the 3D volumetric reconstruction. The current system is capable of obtaining the B-mode 2D images. There needs to be work completed to take these slices and compile them into a 3D reconstructed image. This will include the following: optimizing the step-size as well as exploring different methods to interpolate the data for the most optimal image quality.

Shear Wave Elastography: The shear wave elastography processing needs to go through further development. In order to do this signal processing algorithms should be implemented in order to calculate the values for shear wave velocity. With this information you can use phantoms to mimic the tissue with known stiffness values. In order for this system to be clinically relevant it is extremely important to have qualitative validation that the system can obtain accurate stiffness values.

System Design and Portability: It is very highly recommended that the future team purchases a low cost, newer edition, portable ultrasound system. These ultrasound devices exist at low costs and would give the team flexibility with testing, reduce overlapping needs for shared lab equipment, and get it closer to clinical application. On this note, one of the important goals of this project is decreasing the size of the system and making it a portable ultrasound system. The current system is large, bulky, and heavy. Future design should focus on creating a compact mechanical system, compact component integration, and decreased weight.

Software and User Interface: There is also lots more to improve on the software side of this project. Firstly, there has not been a developed default mode. This will require extensive

testing to determine the most optical presets for a clinician to do a simple default run on the device. Along with this DICOM export needs to be implemented in order to use different softwares and systems in a clinical setting.

Appendix A: Verification Results

Test Verification 1: GUI functionality

Verification Procedure:

1. Launch the GUI and confirm system initialization.
2. Input patient data in the patient intake GUI.
3. Press "Start Scan."
4. Observe the GUI for scan status (active indicator).
5. Press "Stop Scan."
6. Verify that the display reverts to the home screen.

Acceptance Test Data Sheet			
Referenced ATP Paragraph Number: 4.2.3			
Analysis Referenced (for verification by T/A): Test.			
Name of Test: Guided User Interface Functionality.			
Unit Under Test (UUT): Name: Graphical User Interface Module Part Number: S-101			
Results (Pass / Fail): PASS		Date of Test: 02/23/2026	
Recording of Test Measurement: Screen recording of user interaction; log of button response times; screenshots of data display. All processes should work 10 out of 10 tries.	Requirement (SRD, with Tolerances): Displays: Start/Stop Scan shall respond within 5 s and update display status.	Test Equipment Error: n/a	Adjusted Test Limit: n/a
Computations, (Include Analyses Results, if any):			
Signatures: Tester _____ Customer _____			

Test Verification 2: Real-Time B-Mode Image Display

Verification Procedure:

1. Launch the GUI
2. Connect an ultrasound simulation.
3. Begin data transmission.
4. Observe the main display for live B-Mode image updates.
5. Allow the stream to run for 5 minutes to monitor stability.
6. Stop data feed and ensure the GUI returns to the home screen.

Acceptance Test Data Sheet			
Referenced ATP Paragraph Number: 4.2.3			
Analysis Referenced (for verification by T/A): Test.			
Name of Test: Real-Time B-Mode Image Display Functionality			
Unit Under Test (UUT): Name: Graphical User Interface Module Part Number: S-102			
Results (Pass / Fail): PASS		Date of Test: 02/23/2026	
Recording of Test Measurement: Screen capture and system performance log during continuous simulated ultrasound data feed. The process shall run properly 10 out of 10 times.	Requirement (SRD, with Tolerances): The GUI shall display real-time B-mode ultrasound images with a refresh latency ≤ 0.5 s and maintain stable frame updates without freezing or distortion during continuous operation.	Test Equipment Error: n/a	Adjusted Test Limit: n/a
Computations, (Include Analyses Results, if any):			
Signatures: Tester _____ Customer _____			

Test Verification 3: Data Storage and Retrieval

Verification Procedure:

1. Simulate a scan session through the GUI.
2. Save the session using the “Save Scan” option.
3. Ensure that the file name format includes a session ID.
4. Close and reopen the GUI.
5. Retrieve the session using the “Load Scan” function.
6. Confirm the correct image and data load.

Acceptance Test Data Sheet			
Referenced ATP Paragraph Number: 4.2.3			
Analysis Referenced (for verification by T/A): Test.			
Name of Test: Data Storage and Retrieval			
Unit Under Test (UUT): Name: Graphical User Interface Module Part Number: S-103			
Results (Pass / Fail): PASS		Date of Test: 2/23/2026	
Recording of Test Measurement: Screen recording of user interaction; screenshots of data display. Processes shall work properly 10 out of 10 tries.	Requirement (SRD, with Tolerances): The GUI shall export and retrieve the correct imaging data.	Test Equipment Error: n/a	Adjusted Test Limit: n/a
Computations, (Include Analyses Results, if any):			
Signatures: Tester _____ Customer _____			

Test Verification 4: Limit Switch Verification

1. Command the system towards its limit.
2. Observe whether the limit switch triggers and motion stops.
3. Confirm the GUI/serial output reports a limit event.
4. Repeat for the Y-axis.

Acceptance Test Data Sheet			
Referenced ATP Paragraph Number:			
Analysis Referenced (for verification by T/A): Test.			
Name of Test: Limit switch verification.			
Unit Under Test (UUT): Name: Motor assembly Part Number: M-202			
Results (Pass / Fail): PASS		Date of Test: 5/3/2026	
Recording of Test Measurement: Limit switch stops motion on all motors.	Requirement (SRD, with Tolerances): Motion stops immediately upon switch activation, and Teensy reports the correct limit switch.	Test Equipment Error:	Adjusted Test Limit:
Computations, (Include Analyses Results, if any):			
Signatures: Tester _____ Customer _____			

Test Verification 5: Motor Motion Verification

1. Home the system.
2. Command a 10 mm move on the X-axis.
3. Measure the actual displacement using a calliper.
4. Repeat for Y-axis.
5. Command a 90° rotation on the rotation axis.
6. Measure actual rotation using a protractor or marked reference.

Acceptance Test Data Sheet			
Referenced ATP Paragraph Number:			
Analysis Referenced (for verification by T/A): Test.			
Name of Test: Motor motion verification.			
Unit Under Test (UUT): Name: Motor assembly Part Number: M-202			
Results (Pass / Fail): PASS		Date of Test: 02/22/2026	
Recording of Test Measurement: Linear axis moves with ± 0.02 mm accuracy, and the rotation motor moves with $\pm 1^\circ$ accuracy.	Requirement (SRD, with Tolerances): The linear axis moves within ± 0.5 mm. The rotation axis moves within $\pm 2^\circ$.	Test Equipment Error:	Adjusted Test Limit:
Computations, (Include Analyses Results, if any):			
Signatures:			
Tester _____			
Customer _____			

Test Verification 6: Weight Distribution Verification

1. Power on the system, ensure the platform is unloaded, and zero all load-cell readings.
2. Place a calibrated test weights on the platform so that the intended condition is even loading.
3. Record the individual outputs and the computed left/right weight distribution shown in the GUI.
4. Reposition the weight as evenly as possible again and record the readings a second time.
5. For both even trials, compute the percentage difference between left and right sides and compare to the requirement.
6. Confirm that the system correctly indicates a clearly uneven condition when the load is shifted.

Acceptance Test Data Sheet			
Referenced ATP Paragraph Number:			
Analysis Referenced (for verification by T/A): Test.			
Name of Test: WEight Distribution Verification.			
Unit Under Test (UUT): Name: Weight Balance Check Part Number: S-101			
Results (Pass / Fail): Fail		Date of Test: -	
Recording of Test Measurement: Record the total load, and left and right side load.	Requirement: The left-right weight distribution shall be within $\pm 10\%$ of equal loading. The system shall detect and indicate an "uneven" condition when the left-right difference exceeds 10%.	Test Equipment Error:	Adjusted Test Limit:
Computations, (Include Analyses Results, if any):			
Signatures: Tester _____ Customer _____			

Test Verification 7: Probe Orientation Verification

1. Mount the ultrasound probe in the vertical configuration.
2. Place the coupling pad and apply a representative load.
3. Run a full-length scan motion without ultrasound firing, observing probe–pad–platform contact.
4. Visually confirm that the probe maintains continuous contact with the coupling pad across the entire sweep (no gaps or loss of contact).
5. Reconfigure the probe to the horizontal orientation and repeat Steps 2–4.
6. Record locations and conditions where contact is lost or interference occurs.
7. Confirm that both orientations maintain continuous contact over the required scan length.

Acceptance Test Data Sheet			
Referenced ATP Paragraph Number:			
Analysis Referenced (for verification by T/A): Test/Inspection			
Name of Test: Probe orientation verification			
Unit Under Test (UUT): Name: Ultrasound Assembly Part Number: U-201			
Results (Pass / Fail): PASS		Date of Test: 5/3/2026	
Recording of Test Measurement: Vertical Orientation: Contact maintained? Y Horizontal Orientation: Contact maintained? Y	Requirement (SRD, with Tolerances): The probe shall maintain contact with the coupling pad over the full required scan length in both vertical and horizontal orientations.	Test Equipment Error:	Adjusted Test Limit:
Computations, (Include Analyses Results, if any):			
Signatures: Tester _____ Customer _____			

Test Verification 8: SWE Results Verification

1. In MATLAB, load a synthetic shear-wave dataset.
2. Set the reconstruction grid spacing, region of interest, and any required filtering parameters.
3. Execute the SWE MATLAB script to generate a 3D stiffness map.
4. Confirm that the script runs to completion without errors or warnings that indicate instability.
5. Verify that the output stiffness map dimensions and voxel spacing match the configured reconstruction grid.
6. Confirm that stiffness values fall within the expected range.
7. Confirm that the reconstruction meets the specified resolution and stability criteria.

Acceptance Test Data Sheet			
Referenced ATP Paragraph Number:			
Analysis Referenced (for verification by T/A): Test/Analysis			
Name of Test: Probe orientation verification			
Unit Under Test (UUT): Name: Ultrasound Assembly Part Number: SW-003			
Results (Pass / Fail):		Date of Test:	
Recording of Test Measurement: Record the following from the output: Reconstruction grid spacing Output volume size (voxels) Stiffness value range Errors/warnings: Y/N	Requirement (SRD, with Tolerances): The reconstruction software shall generate a 3D stiffness map without numerical failure, and with stiffness values within the expected range for plantar soft tissue.	Test Equipment Error:	Adjusted Test Limit:
Computations, (Include Analyses Results, if any):			
Signatures: Tester _____ Customer _____			

Appendix B: SDD

1.0 Scope

1.1 Identification

- 1.1.1 This Software Design Document (SDD) describes the design of the software system for the Automated Weight-Bearing Ultrasound Foot Scanner (AWBUFS). The software controls motor movement, performs ultrasound imaging procedures, and displays B-Mode, Shear Wave, and volumetric images.

1.2 System Overview

- 1.2.1 The system aims to improve the diagnosis and monitoring of conditions such as progressive collapsing foot deformity (PCFD) by providing images of soft tissue and structural anatomy under weight bearing conditions. The scanner is intended to be portable and clinically valuable for patients and healthcare providers by serving as a non-invasive diagnostic tool.

1.3 Document Overview

- 1.3.1 This SDD includes architectural design, component identification, execution concept, and detailed design for each Computer Software Component (CSC). All these are traced back to system requirements.

2.0 Referenced Documents

- 2.1 System Requirements Document (Section 4.2.5 of the main document)
- 2.2 Software Components Interaction Diagram (DWG - 003)

3.0 CSCI-Wide Design Decisions

3.1 Inputs

- 3.1.1 Path Length in millimeters
- 3.1.2 Step Size in millimeters
- 3.1.3 Script choice, B-Mode or Shearwave
- 3.1.4 Start Scan interaction

3.2 Outputs

- 3.2.1 Motor movement
- 3.2.2 Real-time B-mode images
- 3.2.3 B-mode ultrasound data
- 3.2.4 Real-time Shearwave images
- 3.2.5 Shear wave elastography data

3.3 Behavior in Response to Inputs/Conditions

- 3.3.1 Adjusts total length motor moves
- 3.3.2 Adjusts length motor moves per step

- 3.3.3 Runs either the B-mode scan or the Shearwave scan
- 3.3.4 Starts the scan

3.4 Database/Date File Appearance

- 3.4.1 Raw radiofrequency (RF) data
- 3.4.2 Demodulated in-phase and quadrature (IQ) data
- 3.4.3 Final visualized B-mode image (grayscale)

3.5 Approach to Meeting Safety, Security, and Privacy Requirements

- 3.5.1 Safety
 - 3.5.1.1 Hard motor stop command accessible at all times
- 3.5.2 Security
 - 3.5.2.1 System runs on a dedicated, isolated workstation
 - 3.5.2.2 No network transmission of patient data
- 3.5.3 Privacy
 - 3.5.3.1 Currently no patient identifiers for stored data
 - 3.5.3.2 No direct patient identifiers saved in the imaging process

3.6 Other Design Decisions

- 3.6.1 MATLAB chosen for compatibility with Verasonics system and existing system
- 3.6.2 Modular CSC architecture to allow later replacement of algorithms
- 3.6.3 All CSCs operate within a single MATLAB environment to simplify integration

4.0 CSCI Architectural Design

4.1 Details

- 4.1.1** Component's Identification (Refer to Software Components Interaction Diagram)
 - 4.1.1.1 Movement Algorithm
 - 4.1.1.2 B-Mode Scan Algorithm
 - 4.1.1.3 Shear Wave Scan Algorithm
 - 4.1.1.4 Ultrasound GUI
 - 4.1.1.5 B-Mode Image Display
 - 4.1.1.6 Shear Wave Image Display
- 4.1.2** Static Relationship
 - 4.1.2.1 Refer to Software Components Interaction
- 4.1.3** Purpose
 - 4.1.3.1 Moves motor
 - 4.1.3.2 Dictates how ultrasound operates when conducting B-Mode scan
 - 4.1.3.3 Dictates how ultrasound operates when conducting Shear wave scan
 - 4.1.3.4 Allows user to interact with the system easily to conduct a scan and get desired outputs
 - 4.1.3.5 Allows user to see real-time B-Mode image
 - 4.1.3.6 Allows user to see real-time Shear wave image

4.1.4 Status

- 4.1.4.1 Completed
- 4.1.4.2 Completed
- 4.1.4.3 Completed
- 4.1.4.4 Completed
- 4.1.4.5 Completed
- 4.1.4.6 Completed

4.1.5 Planned Computer Resources

- 4.1.5.1 MATLAB
- 4.1.5.2 Verasonics
- 4.1.5.3 Windows PC workstation
- 4.1.5.4 Motor controller board (Teensy)

4.1.6 Library

- 4.1.6.1 MATLAB Image Processing Toolbox
- 4.1.6.2 MATLAB GUI/AppDesigner Libraries
- 4.1.6.3 Verasonics MATLAB API Packages

4.2 Concept of Execution

- 4.2.1 User launches the MatLab app on the computer
- 4.2.2 User chooses either the B-Mode Imaging script or the Shear Wave Imaging script
 - 4.2.2.1 User starts the scan the same way for both scripts
 - 4.2.2.2 User can and should adjust the gain and voltage for both scripts
 - 4.2.2.3 Depending on the script run, an interface displays the B-Mode image or the Shear wave image
 - 4.2.2.4 Depending on the script run, the delay between motor movement is different
 - 4.2.2.5 Collects data and stores them
- 4.2.3 User can scan again
 - 4.2.3.1 Scanning again takes us back to the first GUI layout for both scripts
- 4.2.4 User exits the script

4.3 Interface Design

- 4.3.1 Identification (Refer to Software Components Interaction Diagram)**
 - 4.3.1.1 Hardware-software interfaces
 - 4.3.1.2 Software-software interfaces
 - 4.3.1.3 User interface
- 4.3.2 Priority**
 - 4.3.2.1 High: motor control, ultrasound firing
 - 4.3.2.2 Medium: real-time reconstruction, GUI updates, data saving
- 4.3.3 Type**
 - 4.3.3.1 MATLAB API (ultrasound)
- 4.3.4 Individual Characteristics**
 - 4.3.4.1 Undetermined

4.3.5 Assembly Characteristics

4.3.5.1 All modules operate in a single MATLAB execution environment

4.3.6 Communication Characteristics

4.3.6.1 Ensure all modules correctly interact with each other, especially the GUI, the existing Verasonics system, and the motor

4.3.7 Protocol Characteristics

4.3.7.1 Verasonics MATLAB parameter file standard

4.3.8 Other Characteristics

4.3.8.1 Real-time performance constraints

5.0 CSCI Detailed Design

5.1 [Unit design decisions (such as algorithms), constraints, language, procedural commands, data handling, logic]

5.2 Will be updated

6.0 Requirement Traceability (Refer to System Requirements)

6.1 Movement Algorithm – Requirement 4.1.4

6.2 B-Mode Scan Algorithm – Requirement 4.1.1, 4.1.3

6.3 Shear Wave Scan Algorithm – Requirement 4.1.1, 4.1.2

6.4 Ultrasound GUI – Requirement 4.3.1

6.5 B-Mode Image Display – Requirement 4.3.2

6.6 Shear Wave Image Display – Requirement 4.3.2

Appendix C: User Manual

*ALL code used for this can be found in the following GitHub Repository:

<https://github.com/tTa-a/Automated-Weight-Bearing-Ultrasound-Foot-Scanner.git>

*Code used in previous years that we did not integrate (weight check system) can be found in the following GitHub Repository:

https://github.com/MartinEncinas/TeamFootPics_24-25

Scan Setup

1. Probe Setup:

- Take the key and place it between the gear and probe holder key slot
- Slide the probe in, fastening the probe along the mounted shoulders. Turn the screw to tighten and keep the probe in place during the motor movements.
 - Make sure to put gel on the probe
 - The probe should be flush against the film layer

2. Gel Setup:

- Place two circular aquaflex gels in the two circular cutouts in the layer on top of the film layer

3. Patient Setup:

- Ensure the back of the patient's foot is in the heel cup
- Ensure the patient is standing on the frame before starting the scan
- Ensure the patient is evenly distributing their weight on the system
- Ensure the patient does not move during the scan

Software/Motor Setup

1. Connect motor usb to pc
2. Turn on motor power (button on box on frame)
3. Check which port the motor connected to (will need to change the code if not COM5)
4. Ensure the motors are in the correct initial positions
 - The limit switches should be working correctly, but if not it's possible to manually readjust the motor or to run the motor control script found in the 2026 GitHub to adjust motor movement
5. Make sure the ultrasound probe is connected to the ultrasound
 - Specifically the L11-4v probe
6. Go to the following directory after freshly opening matlab:
 - C:\Users\Administrator\Documents\Vantage-4.7.6-2206101100_Capstone_2026
 - Type the following into the command window:

- activate
 - Press enter
7. **For the B-mode scan**, click on the following file in the directory we navigated to:
- Cap_Widebeam_Bmode_Motor_022226.m
 - Press the Run button at the top of the file to start the B-mode scan
 - On the UI that pops up, press Start Scan
 - WARNING: make sure the file directory info at the top of the screen (not the editor) still says
“C:\Users\Administrator\Documents\Vantage-4.7.6-2206101100_Capstone_2026” when running the script
 - **WARNING:** make sure in **line 31**, the MOTOR_SERIAL = serialport has the correct port. The current code as COM5, but if it’s plugged into a different port just replace COM5 with what it is actually connected to
 - **WARNING:** in **line 27**, make sure the userMotorFlag = 1 to make the motor move. If you want to run this without moving the motor, change the 1 to 0
 - **WARNING:** in **line 28**, make sure the saveIQFlag = 1 to save the images from a scan. If you want to run the scan without saving any data, change the 1 to 0
 - **To change scan distance or step sizes:**
 - Change the values in **line 38 & 39**
 - **To find the saved data:**
 - **Line 24** shows the savepath, ie where the data is stored
 - c:\CAP26\data\
 - The data will be saved using the date the data was collected
8. **For the Shearwave Elastography Scan**, click on the following file in the directory we navigated to:
- SetUpL11_4_SWI_MOT_RW2d.m
 - **WARNING:** don’t mix it up with SetUpL11_4_SWI_MOT_RW2b.m
 - We want to run the one that ends with D as in DOG
 - Press the Run button at the top of the file to start the SWE-mode scan
 - On the UI that pops up:
 - Press simulation to switch from a simulation to an actual scan
 - Increase the gain slider to get a brighter image
 - Press SWI On to start the scan
 - Increase the gain slider and the voltage slider to get a brighter image
 - **WARNING:** make sure the file directory info at the top of the screen (not the editor) still says
“C:\Users\Administrator\Documents\Vantage-4.7.6-2206101100_Capstone_2026” when running the script

- **WARNING:** make sure in **line 42**, the MOTOR_SERIAL = serialport has the correct port. The current code as COM5, but if it's plugged into a different port just replace COM5 with what it is actually connected to
- **WARNING:** in **line 35**, make sure the userMotorFlag = 1 to make the motor move. If you want to run this without moving the motor, change the 1 to 0
- **WARNING:** in **line 36**, make sure the saveIQFlag = 1 to save the images from a scan. If you want to run the scan without saving any data, change the 1 to 0
- **To change scan distance or step sizes:**
 - Change the values in **line 49 & 50**
- **To find the saved data:**
 - **Line 31** shows the savepath, ie where the data is stored
 - c:\CAP26\data\
 - The data will be saved using the date the data was collected

*Where to find the motor logic scripts besides in the GitHub?

- It's in
 "C:\Users\Administrator\Documents\Vantage-4.7.6-2206101100_Capstone_2026"
- The file names:
 - motor_controls.mlapp
 - motormove.m
 - motorpos.m